International Soil and Water Conservation Research 13 (2025) 600–614

Contents lists available at ScienceDirect

International Soil and Water Conservation Research

journal homepage: www.elsevier.com/locate/iswcr

Original Research Article

Development of web-based decision support tool for rainfall erosivity estimation using both high-resolution rainfall data and simplified models

Sinae Kim [a], Seung-Oh Hur [b], Jihye Kwak [a], Jihye Kim [c], Moon-Seong Kang [d, *]

[a] Department of Rural Systems Engineering, Seoul National University, Seoul, 08826, Republic of Korea
[b] National Institute of Agricultural Sciences, Rural Development Administration, Jeonju, 55365, Republic of Korea
[c] Research Institute of Agriculture and Life Science, Seoul National University, Seoul, 08826, Republic of Korea
[d] Department of Rural Systems Engineering, Research Institute for Agriculture and Life Science, and Institutes of Green Bio Science and Technology, Seoul National University, Seoul, 08826, Republic of Korea

A R T I C L E I N F O

Article history:
Received 26 September 2024
Received in revised form
16 March 2025
Accepted 16 March 2025
Available online 17 March 2025

Keywords:
Soil erosion
USLE
Minute-interval rainfall
Rainfall erosivity
Regression model

A B S T R A C T

Soil erosion is a significant global problem that has far-reaching effects on agricultural productivity, environmental health, and ecosystem stability. The rainfall erosivity factor (R-factor) used in the Universal Soil Loss Equation (USLE) is a key parameter for predicting soil erosion. However, its accurate estimation is difficult owing to the complexities of high-resolution rainfall data and limitations of simplified models. This study addressed these challenges by introducing several key innovations. We developed a precise algorithm for calculating the R-factor using minute-interval rainfall data to effectively capture the necessary temporal resolution for assessing the impacts of extreme rainfall events. This advancement allows for accurate R-factor estimation, thereby overcoming the complexities associated with high-resolution data processing. In addition, we established a comprehensive rainfall erosivity database across South Korea based on 24 years of minute-interval rainfall data. We then derived an optimal regression model for estimating monthly rainfall erosivity from daily precipitation data, achieving high accuracy ($R^2 = 0.87$) by effectively accounting for extreme rainfall events. These efforts culminated in the development of the Web-based Rainfall Erosivity Calculation (WREC) tool, which integrates a database, a rainfall erosivity calculation algorithm, and a simple estimation model. The user-friendly interface of the WREC tool offers a versatile platform for calculating rainfall erosivity, supporting practical applications, and assessing future climate change impacts. Expanding the WREC tool globally and adapting regression models to local contexts will enhance our ability to manage soil erosion and promote sustainable land and water management practices.

1. Introduction

Soil erosion by water is a significant global challenge in the agricultural and environmental sectors that can lead to diminished agricultural productivity and various social and environmental problems, including nutrient runoff, decreased river discharge capacity, increased landslide incidence, and elevated turbidity in rivers and lakes (Hwang et al., 2016; Meena et al., 2017; Mullan, 2013; Rickson, 2014). Furthermore, soil erosion adversely affects ecological aspects, contributing to water pollution, biodiversity loss, and landscape and recreational activity impairment (Panagos et al., 2015). Notably, recent changes in rainfall patterns attributed to climate change have led to an increase in the intensity and frequency of extreme rainfall events (IPCC, 2022). Alterations in rainfall erosivity owing to climate change can increase the risk of accelerated soil erosion (Borrelli et al., 2020; Panagos et al., 2022).

* Corresponding author.
E-mail addresses: ksinae0519@snu.ac.kr (S. Kim), soilssohur@korea.kr (S.-O. Hur), dbk02300@snu.ac.kr (J. Kwak), jewisdom@naver.com (J. Kim), mskang@snu.ac.kr (M.-S. Kang).

Peer review under the responsibility of International Research and Training Center on Erosion and Sedimentation, the China Water and Power Press, and China Institute of Water Resources and Hydropower Research.

https://doi.org/10.1016/j.iswcr.2025.03.003
2095-6339/

Given that soil erosion is a major global concern, soil protection and restoration must be considered through the accurate prediction of soil loss, which is essential for preventing erosion and developing appropriate management strategies (Jang et al., 2015). Over the past few decades, various empirical and physical-based models have been developed to predict soil loss (Borrelli et al., 2021; De Vente & Poesen, 2005). Among these, the Revised Universal Soil Loss Equation (RUSLE) is the most renowned and widely applied model for soil erosion prediction globally. The RUSLE model is an empirical model that can predict the long-term average annual rate of soil erosion on slopes using rainfall, soil type, topography, crop systems, and management practices (Mondal et al., 2016; Pandey et al., 2021; Prasannakumar et al., 2011). Despite its limitations, such as not simulating soil deposition (e.g., sedimentation) and insufficient measured data for all required situations and scenarios, the RUSLE model remains popular owing to its high flexibility, data accessibility, and simplicity of parameterization (Alewell et al., 2019). In particular, the RUSLE model has been extensively applied to soil erosion prediction in South Korea, supported by a wealth of prior research that has established a robust dataset of various environmental conditions and regional parameters. Consequently, this facilitated the comparison and evaluation of the soil erosion estimation results and their applicability.

The primary drivers of soil erosion include the splashing effect of raindrops hitting the ground and scouring effect of surface runoff (Yin et al., 2017). Consequently, among the five individual factors in the (R)USLE formula, the rainfall erosivity factor (R-factor) is a vital component because it encompasses the erosion caused by the kinetic energy of rainfall and surface runoff (Wischmeier & Smith, 1978). The accuracy of the R-factor significantly influences the accuracy of the soil loss estimation (Wang et al., 2024). In South Korea, which has a monsoon climate, significant variations in summer rainfall have raised concerns regarding increased soil erosion due to intense summer rainstorms. Additionally, with a high proportion of mountainous terrain and many upland fields situated on slopes, accounting for over 50 % of the total soil loss, the risk of soil loss due to rainfall and runoff is elevated. Therefore, the R-factor, which is more variable than other factors, is essential for accurate soil erosion estimation and prediction (Kang et al., 2009; Lee et al., 2011).

Rainfall erosivity is calculated by summing the products of the kinetic energy of rainfall and the maximum 30-min intensity of each storm event (Brown & Foster, 1987; Wischmeier & Smith, 1978). High-resolution rainfall data are required to accurately distinguish between rainfall events and determine the maximum 30-min intensity. At least 20 years of rainfall data are needed to estimate the long-term average annual R-factor (Renard & Freimund, 1994). However, obtaining subhourly rainfall data over 20 years is often challenging. Even when such data are available, the process of collecting high-resolution meteorological data, classifying rainfall events, and calculating the rainfall energy and maximum 30-min intensity for each event is complex and labor-intensive (Lee & Heo, 2011). Consequently, manual calculation of the R-factor is time-consuming and error-prone, which can compromise the accuracy of the results (Cardoso et al., 2020). Owing to these difficulties, researchers have often relied on existing rainfall erosivity values from prior studies rather than recalculating them.

In South Korea, the rainfall erosivity values calculated by Jung et al. (1983) and Park et al. (2000) using rainfall data before 2000 are used by the Ministry of the Environment as standards for soil erosion estimation. Despite the numerous studies aimed at recalculating nationwide rainfall erosivity since the 1980s (Jung et al., 1983; Kang et al., 2003, 2021; Park et al., 2000, 2011), most have relied on hourly rainfall data because of the limitations in obtaining sub-hourly data. Some studies, including those by Lee and Lin (2015), Risal et al. (2016), and Shin et al. (2019), which utilized 10- or 5-min data, had datasets that were generally limited to <20 years. With the recent availability of extensive long-term minute-interval meteorological data in South Korea, the accuracy of rainfall erosivity estimates can be considerably enhanced. However, calculating long-term rainfall erosivity from such high-resolution data is complex and labor-intensive. To address this challenge, an efficient algorithm coupled with a user-friendly web-based tool must be developed.

To address the challenges of obtaining high-resolution rainfall data, various simplified estimation models have been proposed to calculate rainfall erosivity using available rainfall data. The most common form of these models is regression-based and estimates annual rainfall erosivity from annual precipitation, which has been the focus of numerous studies across different countries (El-Swaify et al., 1985; Lo et al., 1985; Renard & Freimund, 1994; Yu & Rosewell, 1996; Lin et al., 2002; Sepaskhah & Sarkhosh, 2005; Bonilla & Vidal, 2011; Lee & Heo, 2011; Yin et al., 2015). However, annual-scale estimates may not adequately capture seasonal variations in erosivity, which are particularly relevant for soil conservation planning and short-term risk assessments (Marcinkowski & Bagdasaryan, 2024; Wang et al., 2023).

Recently, researchers have also developed regression equations to estimate rainfall erosivity from daily and monthly precipitation data, as well as from the maximum daily and hourly precipitation. Among these, monthly rainfall erosivity models have been highlighted as a more reliable alternative because they provide improved accuracy over daily models while maintaining greater temporal resolution than annual models (Lee & Lin, 2015). Monthly-scale erosivity estimates are particularly useful in seasonal soil erosion modeling and have been applied in studies examining intra-annual variations in soil loss, sediment yield, and land management effectiveness (Cao et al., 2023; Johannsen et al., 2022; Polykretis et al., 2020; Schmidt et al., 2019). Moreover, because RUSLE calculate long-term average soil loss, monthly R-factor values can be integrated into modified versions of these models to estimate seasonal soil erosion risk. Given the importance of capturing seasonal erosivity changes, it is necessary to determine the optimal parameter combinations for estimating monthly rainfall erosivity using daily precipitation data. Developing robust regression equations for monthly R-factor estimation will enhance soil erosion modeling and facilitate climate change impact assessment.

In South Korea, the variability in rainfall erosivity is significant across different regions and periods; this variability is expected to increase further because of recent changes in rainfall patterns associated with climate change (Shin et al., 2019). Predicting future changes in rainfall erosivity under different climate change scenarios is crucial for identifying areas at risk of soil erosion. However, the daily resolution of climate change scenarios provided by Global Climate Models renders the application of minute-interval data impractical. Incorporating a simple estimation model for monthly rainfall erosivity using daily data into a web-based tool would facilitate the estimation of future changes in rainfall erosivity due to climate change and enable the estimation of rainfall erosivity in regions where minute-interval meteorological data are not available.

This study aimed to provide a decision support tool for estimating rainfall erosivity based on available rainfall data by developing a Web-based Rainfall Erosivity Calculation (WREC) tool that integrates a rainfall erosivity calculation algorithm and simplified models. To achieve this, the objectives of this study were to (a) develop an algorithm for calculating rainfall erosivity based on minute-interval rainfall data, (b) create a database of rainfall erosivity for 75 meteorological stations across South Korea using 24 years of minute-interval rainfall data, and (c) derive an optimal regression model for monthly rainfall erosivity estimation. This tool would streamline the rainfall erosivity calculation process, making it easy for users to obtain accurate and up-to-date rainfall erosivity values, thereby contributing to more reliable soil erosion assessments and environmental planning.

2. Material and methods

2.1. Study area

South Korea (33–43°N, 124–132°E) is a peninsular nation in the northwest Pacific region. It covers a total land area of approximately 223,000 km^2, with approximately 70 % of this area comprising mountainous terrain. Its northern and eastern regions are characterized by high mountain ranges, whereas the southern and western regions are predominantly composed of lower mountain ranges. The country experiences a monsoon climate with four distinct seasons and has an average annual precipitation of 1,306.3 mm (1991–2020). Notably, precipitation is concentrated during summer, accounting for 54 % of the total annual precipitation. Natural disasters, particularly typhoons and heavy rainfall, cause significant nationwide damage (MOIS, 2023), raising concerns regarding increased soil erosion due to intense rainfall. The location of the study area, a digital elevation model, and positions of the 75 Automated Surface Observing System (ASOS) weather stations in South Korea are shown in Fig. 1.

According to soil erosion grades from the Organization for Economic Co-operation and Development (2008), the average annual soil loss in the country is approximately 33 tons/ha and is classified as "very severe." Areas with soil losses exceeding 50 tons/ha account for 20 % of the total land area (ME, 2013). In recent years, substantial soil erosion has been observed, particularly in highland agricultural areas cultivated on slopes. According to the Intergovernmental Panel on Climate Change (IPCC), the future (2080–2100) maximum daily precipitation in the Republic of Korea is projected to increase by 20–37 % of the current level (125.7 mm) (IPCC, 2022), indicating that soil erosion due to rainfall is expected to intensify as a result of climate change.

2.2. Rainfall data

In this study, 1-min interval and daily rainfall data for 2000–2023 from the 75 ASOS weather observation stations across South Korea were provided by the Korea Meteorological Administration (KMA). Table 1 presents the periods of data availability at 1-min intervals along with the rainfall characteristics, including average precipitation, standard deviation (SD), and coefficient of variation (CV) for each of the 75 weather stations. The national average annual precipitation was 1,354.8 mm, with an average annual precipitation standard deviation of 323.9 mm and a coefficient of variation of 24 %. The average annual precipitation levels ranged from a minimum of 818.5 mm to a maximum of 2,035.3 mm, reflecting considerable regional variability in annual precipitation across the country.

2.3. Calculation of rainfall erosivity

2.3.1. Rainfall erosivity calculation

The rainfall erosivity quantifies the degree of soil erosion caused by rainfall and is defined as the product of the kinetic energy of a rainfall event and the maximum 30-min rainfall intensity within

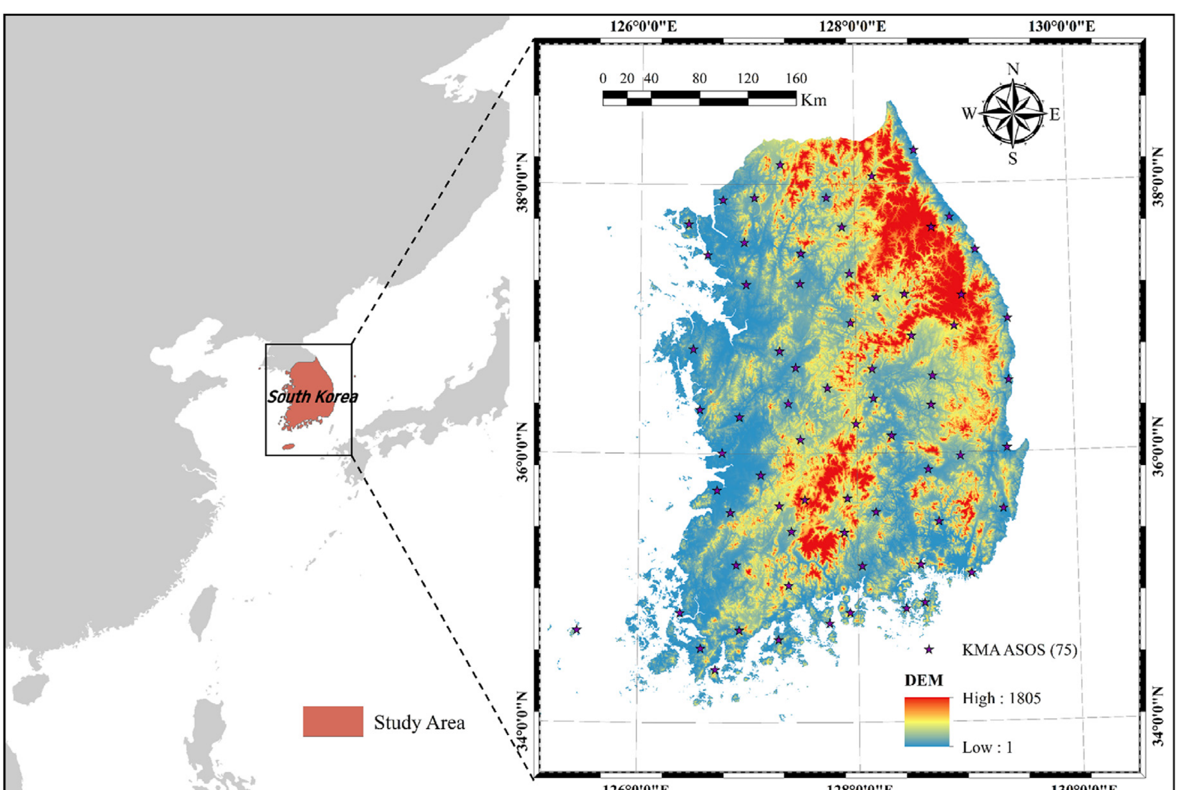

Fig. 1. Location and digital elevation map of the study area.

Table 1
Information on weather stations with 1-min interval rainfall data and rainfall characteristics (average annual rainfall, annual rainfall standard deviation, coefficient of variation).

No.	Station code	Station name	Data period	Latitude	Longitude	Average (mm)	SD (mm)	CV
1	90	Sokcho	2000–2023	38°15′N	128°33′E	1436.3	267.4	0.186
2	95	Cheorwon	2000–2023	38°8′5N	127°18′E	1382.1	292.4	0.212
3	98	Dongducheon	2000–2023	37°54′N	127°3′3E	1408.2	355.9	0.253
4	99	Paju	2002–2023	37°53′N	126°45′E	1310.3	350.7	0.268
5	100	Daegwallyeong	2001–2023	37°40′N	128°43′E	1512.7	484.1	0.320
6	101	Chuncheon	2000–2023	37°54′N	127°44′E	1360.9	327.1	0.240
7	102	Baengnyeongdo	2001–2023	37°58′N	124°42′E	818.5	239.6	0.293
8	105	Gangneung	2000–2023	37°45′N	128°53′E	1451.3	326.5	0.225
9	106	Donghae	2000–2023	37°30′N	129°7′2E	1295.2	303.3	0.234
10	108	Seoul	2000–2023	37°34′N	126°57′E	1416.3	356.7	0.252
11	112	Incheon	2000–2023	37°28′N	126°37′E	1219.6	288.7	0.237
12	114	Wonju	2000–2023	37°20′N	127°56′E	1296.5	340.5	0.263
13	115	Ulleungdo	2000–2023	37°28′N	130°53′E	1555.5	315.3	0.203
14	119	Suwon	2000–2023	37°15′N	126°58′E	1338.7	290.2	0.217
15	121	Yeongwol	2000–2023	37°10′N	128°27′E	1217.3	319.2	0.262
16	127	Chungju	2002–2023	36°58′N	127°57′E	1263.3	326.2	0.258
17	129	Seosan	2000–2023	36°46′N	126°29′E	2022.4	364.7	0.180
18	130	Uljin	2000–2023	36°59′N	129°24′E	1197.0	273.4	0.228
19	131	Cheongju	2000–2023	36°38′N	127°26′E	1244.7	301.8	0.242
20	133	Daejeon	2000–2023	36°22′N	127°22′E	1357.3	338.3	0.249
21	135	Chupungnyeong	2000–2023	36°13′N	127°59′E	1208.0	269.6	0.223
22	136	Andong	2000–2023	36°34′N	128°42′E	1049.6	228.0	0.217
23	137	Sangju	2002–2023	36°24′N	128°9′2E	1190.2	268.6	0.226
24	138	Pohang	2000–2023	36°1′5N	129°22′E	1186.5	290.7	0.245
25	140	Gunsan	2000–2023	36°0′1N	126°45′E	1299.1	348.9	0.269
26	143	Daegu	2000–2023	35°52′N	128°39′E	1075.3	263.0	0.245
27	146	Jeonju	2000–2023	35°50′N	127°7′1E	1322.8	305.6	0.231
28	152	Ulsan	2000–2023	35°34′N	129°20′E	1263.8	292.9	0.232
29	155	Changwon	2000–2023	35°10′N	128°34′E	1534.6	345.0	0.225
30	156	Gwangju	2000–2023	35°10′N	126°53′E	1416.5	340.9	0.241
31	159	Busan	2000–2023	35°6′1N	129°1′5E	1580.1	393.9	0.249
32	162	Tongyeong	2000–2023	34°50′N	128°26′E	1567.8	338.2	0.216
33	165	Mokpo	2000–2023	34°49′N	126°22′E	1201.6	254.8	0.212
34	168	Yeosu	2000–2023	34°44′N	127°44′E	1470.4	310.9	0.211
35	169	Heuksando	2000–2023	34°41′N	125°27′E	1107.5	263.3	0.238
36	170	Wando	2000–2023	34°23′N	126°42′E	1546.0	340.0	0.220
37	174	Suncheon	2000–2023	35°1′1N	127°22′E	1601.9	375.0	0.234
38	184	Jeju	2000–2023	33°30′N	126°31′E	1487.8	383.7	0.258
39	185	Gosan	2000–2023	33°17′N	126°9′4E	1220.7	230.2	0.189
40	188	Seongsan	2000–2023	33°23′N	126°52′E	2035.3	371.1	0.182
41	189	Seogwipo	2000–2023	33°14′N	126°33′E	1979.7	417.8	0.211
42	192	Jinju	2000–2023	35°9′4N	128°2′2E	1550.9	386.4	0.249
43	201	Ganghwa	2000–2023	37°42′N	126°26′E	1248.7	326.9	0.262
44	202	Yangpyeong	2000–2023	37°29′N	127°29′E	1408.2	393.4	0.279
45	203	Icheon	2000–2023	37°15′N	127°29′E	1310.7	314.1	0.240
46	211	Inje	2000–2023	38°3′3N	128°10′E	1222.4	293.2	0.240
47	212	Hongcheon	2000–2023	37°41′N	127°52′E	1334.6	361.2	0.271
48	216	Taebaek	2000–2023	37°10′N	128°59′E	1302.6	317.7	0.244
49	221	Jecheon	2000–2023	37°9′3N	128°11′E	1382.5	391.2	0.283
50	226	Boeun	2000–2023	36°29′N	127°44′E	1326.3	326.6	0.246
51	232	Cheonan	2000–2023	36°45′N	127°17′E	1214.3	280.0	0.231
52	235	Boryeong	2000–2023	36°19′N	126°33′E	1198.4	294.4	0.246
53	236	Buyeo	2000–2023	36°16′N	126°55′E	1339.1	354.3	0.265
54	238	Geumsan	2000–2023	36°6′2N	127°28′E	1309.5	303.5	0.232
55	243	Buan	2000–2023	35°43′N	126°42′E	1287.4	320.2	0.249
56	244	Imsil	2000–2023	35°36′N	127°17′E	1380.5	320.5	0.232
57	245	Jeongeup	2000–2023	35°33′N	126°50′E	1353.8	293.1	0.216
58	247	Namwon	2000–2023	35°25′N	127°23′E	1382.6	340.5	0.246
59	248	Jangsu	2000–2023	35°39′N	127°31′E	1513.1	366.2	0.242
60	260	Jangheung	2000–2023	34°41′N	126°55′E	1509.9	335.4	0.222
61	261	Haenam	2000–2023	34°33′N	126°34′E	1311.9	266.2	0.203
62	262	Goheung	2000–2023	34°37′N	127°16′E	1473.0	319.8	0.217
63	271	Bonghwa	2000–2023	36°56′N	128°54′E	1198.0	288.2	0.241
64	272	Yeongju	2000–2023	36°52′N	128°31′E	1346.2	339.0	0.252
65	273	Mungyeong	2000–2023	36°37′N	128°8′5E	1326.5	303.3	0.229
66	277	Yeongdeok	2000–2023	36°32′N	129°24′E	1125.4	261.2	0.232
67	278	Uiseong	2000–2023	36°21′N	128°41′E	1032.4	312.5	0.303
68	279	Gumi	2000–2023	36°7′5N	128°19′E	1151.7	309.7	0.269
69	281	Yeongcheon	2000–2023	35°58′N	128°57′E	1069.5	237.8	0.222
70	284	Geochang	2000–2023	35°40′N	127°54′E	1302.6	354.7	0.272
71	285	Hapcheon	2000–2023	35°33′N	128°10′E	1345.5	322.5	0.240
72	288	Miryang	2000–2023	35°29′N	128°44′E	1221.3	300.9	0.246

(*continued on next page*)

Table 1 (*continued*)

No.	Station code	Station name	Data period	Latitude	Longitude	Average (mm)	SD (mm)	CV
73	289	Sancheong	2000–2023	35°24′N	127°52′E	1573.9	412.6	0.262
74	294	Geoje	2000–2023	34°53′N	128°36′E	1936.5	468.6	0.242
75	295	Namhae	2000–2023	34°48′N	127°55′E	1934.9	426.8	0.221
Average						1354.8	323.9	0.240

*Note: SD, standard deviation; CV, coefficient of variation.

the storm event. To estimate the annual average rainfall erosivity, the effective rainfall events that caused soil erosion were classified. Wischmeier and Smith (1978) considered a rainfall event as an event during which the total rainfall was 12.7 mm or more and the interval between rainfall events was within 6 h. Even if the total rainfall was <12.7 mm, if the rainfall within 15 min exceeded 6.25 mm, it was considered as a rainfall event that could cause soil erosion. Effective rainfall events were defined as those with a total precipitation of ≥12.7 mm or rainfall of ≥6.25 mm within 15 min.

After classifying effective rainfall events, the procedure for calculating the R-factor in the USLE was as follows: extraction of the maximum 30-min rainfall intensity for each rainfall event, calculation of the total storm kinetic energy, and determination of the rainfall erosivity for individual storm events by multiplying the maximum 30-min rainfall intensity by the storm kinetic energy. The formula for estimating the R-factor over a long period is given in Eq. (1).

$$R = \frac{1}{n}\sum_{j=1}^{n}\sum_{i=1}^{m_j} EI_{30} \tag{1}$$

where R is the average annual rainfall erosivity factor (MJ-mm/ha/hr/yr), n is the number of years of data, m_j is the number of erosive rainfall events in year j, and EI_{30} is the rainfall erosivity value of individual storm events (MJ-mm/ha/hr). EI_{30} is calculated using Eq. (2).

$$EI_{30} = E \times I_{30\ max} \tag{2}$$

where E is the total storm kinetic energy of each storm event (MJ/ha) and I_{30} is the maximum 30-min rainfall intensity (mm/hr) within the rainfall event. The total storm kinetic energy, E is calculated using Eq. (3).

$$E = \sum_{r=1}^{p} e_r \cdot v_r \tag{3}$$

where e_r is the unit rainfall energy (MJ/ha/mm) and v_r is the rainfall volume (mm) during time period of r, and p is the number of periods in the rainfall event. The unit of rainfall energy (e_r) used in Eq. (3) is calculated using Eqs. (4) and (5) as follows:

$$e_r = 0.119 + 0.0873\ log_{10} I_r \quad if\ I_r \leq 76 mm/hr \tag{4}$$

$$e_r = 0.283 \quad if\ I_r > 76 mm/hr \tag{5}$$

where I_r is the rainfall intensity during the time interval (mm/hr). If the rainfall intensity is > 76 mm/h, the unit rainfall energy was taken as 0.283 MJ/ha/mm (Renard & Freimund, 1994; Wischmeier & Smith, 1978).

The calculation of the R-factor requires rainfall data observed at intervals of at least 5 min and involves complex processes. Consequently, researchers often rely on R-factor values and iso-erodent maps provided in prior studies instead of calculating them directly. In South Korea, the most commonly used rainfall erosivity values for meteorological stations are based on research by Jung et al. (1983) and Park et al. (2000). However, these values, derived from rainfall data collected before 2000, are considered outdated and limited by the insufficient duration of the data period used.

Therefore, we developed an algorithm for quickly estimating the R-factor with minimal computational errors. Using this algorithm, R-factors for meteorological stations across South Korea were updated based on long-term high-resolution rainfall data. The algorithm uses daily cumulative 1-min interval rainfall data (the basic format of minute-by-minute rainfall data provided by the KMA) as the input and calculates the R-factor according to the USLE method. It effectively handles missing values and provides results for the maximum 30-min rainfall intensity, total kinetic energy, and rainfall erosivity for each rainfall event. A schematic of the algorithm is shown in Fig. 2.

2.3.2. Simple estimation models

The USLE method for calculating the R-factor requires high-resolution rainfall data and involves intricate and labor-intensive computational procedures. To address these challenges, recent studies have proposed simple regression-based methods for estimating the R-factor using annual, monthly, and daily rainfall data (Wang et al., 2024). These simplified estimation models have also been employed in various studies to estimate the R-factor for future periods, even with lower temporal resolution weather data (Grillakis et al., 2020; dos Santos et al., 2022; Xu et al., 2023; Chen et al., 2024).

In this study, we developed a regression equation to estimate monthly rainfall erosivity from daily rainfall data. This approach leverages a higher temporal resolution of daily data to provide a more detailed analysis of extreme rainfall events. To develop the regression equation, we utilized monthly rainfall erosivity data estimated from 1-min interval rainfall data following the R-factor calculation algorithm described in Section 2.3.1.

As detailed in Table 2, we considered the variables used in prior research to estimate monthly rainfall erosivity as potential predictors. All variables were calculated using daily rainfall data. To identify the optimal regression model, we employed the Curve Expert Professional (v.2.7.3) program, which facilitates fitting and comparing over 90 linear and nonlinear regression models to select the best fit. Data from 60 of the 75 weather observation stations were randomly selected for the calibration set, whereas the remaining 15 stations were used for validation. We determined the optimal regression equation using data from the 60 stations, and verified the accuracy of each regression model using data from the 15 validation stations.

2.4. WREC tool development

With the accumulation of long-term weather data, rapid and accurate estimation of the R-factor remains a significant challenge in recent research despite various researchers having developed R-factor estimation programs and tools (Araujo et al., 2018, pp. 1–8; Cardoso et al., 2020; Moreira et al., 2008; Risal et al., 2016). However, these tools often have limitations such as requiring specific time intervals for input dataset, lacking flexibility in input dataset formats, and necessitating the installation of separate programs or packages, which can impede user accessibility and maintenance.

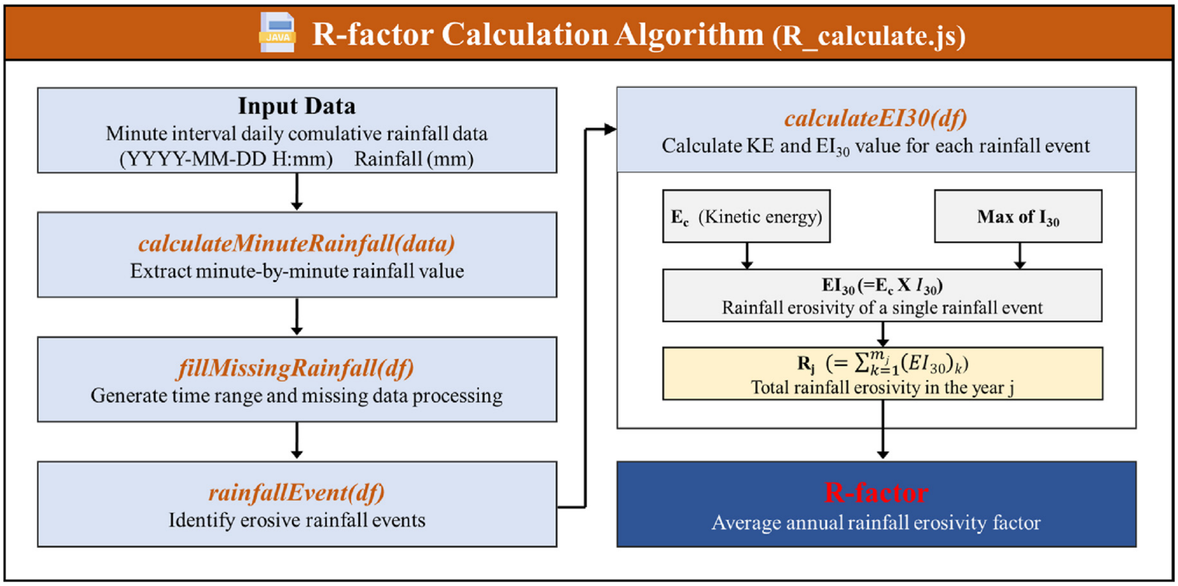

Fig. 2. Schematic diagram of the rainfall erosivity factor calculation algorithm.

Table 2
Parameters predominantly used in regression models for monthly rainfall erosivity prediction.

Parameter	Parameter meaning	References
P_m	Monthly rainfall amount (mm)	Huang et al. (1992); Wu (1994)
MFI_m	MFI (Modified Fournier Index) of month (P_m^2/P_{annual})	Arnoldus (1977); Pontes et al. (2017)
P_{md}	Monthly maximum daily rainfall amount (mm)	Diodato (2006); Grauso et al. (2010)
P_{10}	Monthly rainfall amount for days rainfall≥10 mm (mm)	Diodato (2004); De Santos Loureiro and de Azevedo Coutinho (2001); Capolongo et al. (2008)
P_{10d}	Monthly number of days with rainfall ≥10 mm (days)	
M	Order of month	Diodato and Bellocchi (2007); Risal et al. (2016)

*Note: P_{annual}, annual rainfall amount (mm).

To address these challenges, we developed the WREC tool, which supports both calculations based on 1-min interval rainfall data using the USLE method and estimations based on daily rainfall data using a simplified method. As a web-based platform, the WREC tool ensures accessibility to any location, facilitating ease of use for a wide range of users.

The frontend of the WREC program was developed using HyperText Markup Language (HTML), Cascading Style Sheets (CSS), and JavaScript, allowing for various functions such as user input processing and data visualization. The backend was developed using Node.js, a JavaScript runtime environment that employs an event-driven asynchronous I/O model, along with the Express framework.

The tool consists of three main modules.

1. **DB-based module:** This module allows users to view rainfall erosivity values for ASOS weather stations across South Korea using a pre-existing nationwide database without requiring any additional input data.
2. **USLE-based module:** Users can upload 1-min interval rainfall data to this module. The system then calculates the monthly and annual rainfall erosivity values according to the USLE method, utilizing the algorithm illustrated in Fig. 2.
3. **Regression-based module:** In cases where high-resolution weather data are unavailable, users can input daily rainfall data. The system provides estimates based on a user-selected regression model.

The WREC tool can accommodate various methods for calculating R-factors depending on the available format of rainfall data, offering users the flexibility to select the most suitable analytical approach for their data. A schematic of the WREC tool web interface and backend processing is shown in Fig. 3.

3. Results

3.1. Rainfall erosivity map in South Korea

Using 1-min interval weather data, we calculated the R-factors for 75 ASOS weather stations from 2000 to 2023 following the USLE method. Fig. 4 shows the iso-erodent map of South Korea, created using the Kriging spatial interpolation technique based on the annual average rainfall erosivity values from 2000 to 2023. The Kriging method accounts for spatial autocorrelation by utilizing a semivariogram model and offers significant advantages over non-geostatistical methods such as Inverse Distance Weighting (IDW) and spline interpolation. Unlike purely distance-based approaches, Kriging estimates unknown values based on the weighted average of neighboring observations, considering both distance and spatial dependence. This allows for a more accurate representation of localized fluctuations and the underlying spatial structure of rainfall erosivity. Given these advantages and the strong spatial gradients in rainfall erosivity, Kriging was selected as the preferred interpolation method in this study to improve spatial accuracy and reliability.

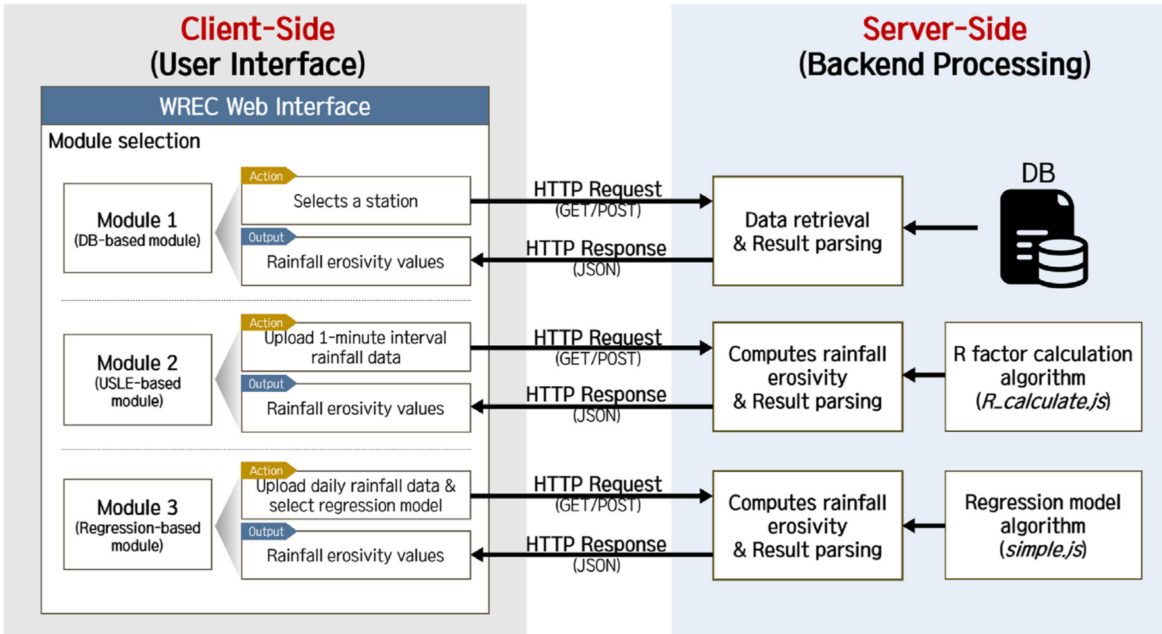

Fig. 3. Schematic diagram of web interface and backend processing of the WREC tool.

The map highlights significant regional variations in rainfall erosivity. Generally, the southern regions experienced higher rainfall erosivity than the central areas, with notably elevated values observed in the northwestern and southeastern regions. The national average annual rainfall erosivity was 5,252 MJ-mm/ha/hr/yr. The highest recorded value was 10,336 MJ-mm/ha/hr/yr at the Seongsan (188) weather station, whereas the lowest was 2,495 MJ-mm/ha/hr/yr at the Baengnyeongdo (102) weather station. Notably, the weather stations with the highest and lowest R-factors corresponded to those with the highest and lowest annual precipitation, respectively.

The intra-annual distribution of erosion risks was analyzed by examining the characteristics of monthly rainfall erosivity and erosivity density (which represents rainfall erosivity per unit of precipitation). Erosivity density is an index that aids in understanding erosion processes, as it describes the interaction between the kinetic energy of raindrops and the soil surface. The average monthly rainfall erosivity and erosivity density for each year from 2000 to 2023 are shown in Fig. 5.

Fig. 5(a) shows that rainfall erosivity values peaked during the flood season (June to September), due to frequent typhoons and intense rainfall events. On average, the rainfall erosivity during this period accounted for 84.9 % of the annual rainfall erosivity, with values particularly concentrated in July and August. The highest average monthly rainfall erosivity was observed in July, with an average value of 1658 MJ-mm/ha/hr/mo, while the lowest was in January, with an average of 18 MJ-mm/ha/hr/mo. The CV of annual rainfall erosivity was 51 %, indicating substantial inter-annual variability.

In contrast, Fig. 5(b) shows that erosivity density during the flood season accounted for 68 % of the total annual erosivity density. Unlike rainfall erosivity values, erosivity density was consistently high during these months, resulting in lower intra-annual variability. The highest average monthly erosivity density was recorded in August at 4.7 MJ/ha/hr/mo, whereas the lowest was in December, with a value of 0.3 MJ/ha/hr/mo. These findings demonstrate that erosivity density does not increase or decrease in direct proportion to rainfall or rainfall erosivity values. Instead, the combined effects of rainfall energy, intensity, and volume influence erosion density, leading to variations in the erosive power per unit of rainfall. The CV of annual erosivity density was 35 %, suggesting that it is less influenced by inter-annual variability compared to rainfall erosivity values.

3.2. Derivation of a regression model for estimating monthly rainfall erosivity

To predict monthly rainfall erosivity using daily rainfall data, we selected ten parameter combinations based on the six parameters listed in Table 2 (P_m, MFI_m, P_{md}, P_{10}, P_{10d}, and M), as shown in Table 3. We ensured that there was no multicollinearity among the independent parameters (VIF <5) when more than two parameters were included.

We derived the optimal regression models (R1–R9) for each parameter combination using data from the 60 randomly selected ASOS weather stations, excluding those with station codes 90, 98, 100, 102, 133, 155, 162, 165, 170, 212, 245, 261, 277, 284, and 295. The analysis revealed that the power models (Power and Hoerl) were generally the most accurate in predicting monthly rainfall erosivity, except for models R2, R8, and R10.

The regression equations determined from the calibration set were then used to predict the monthly rainfall erosivity for the validation set consisting of the remaining 15 weather stations. The accuracy of these predictions was assessed (Fig. 6). Among the models, the regression model (R9) using the monthly maximum daily rainfall (P_{md}) and monthly rainfall amount for days >10 mm (P_{10}) as parameters demonstrated the highest predictive accuracy, with the highest R^2 and lowest MAE for both the calibration and validation sets, as illustrated in Fig. 6(a).

Fig. 6(b) presents a scatterplot comparing the actual rainfall erosivity values calculated using the USLE method with those predicted from the R9 regression model. The R^2 was >0.85 for both the calibration and validation sets, indicating that the regression equation derived in this study ($R = 0.280 \cdot P_{10}{}^{1.075} \cdot P_{md}{}^{0.545}$) could predict monthly rainfall erosivity with high accuracy.

Additionally, Fig. 7 presents a contour plot illustrating the

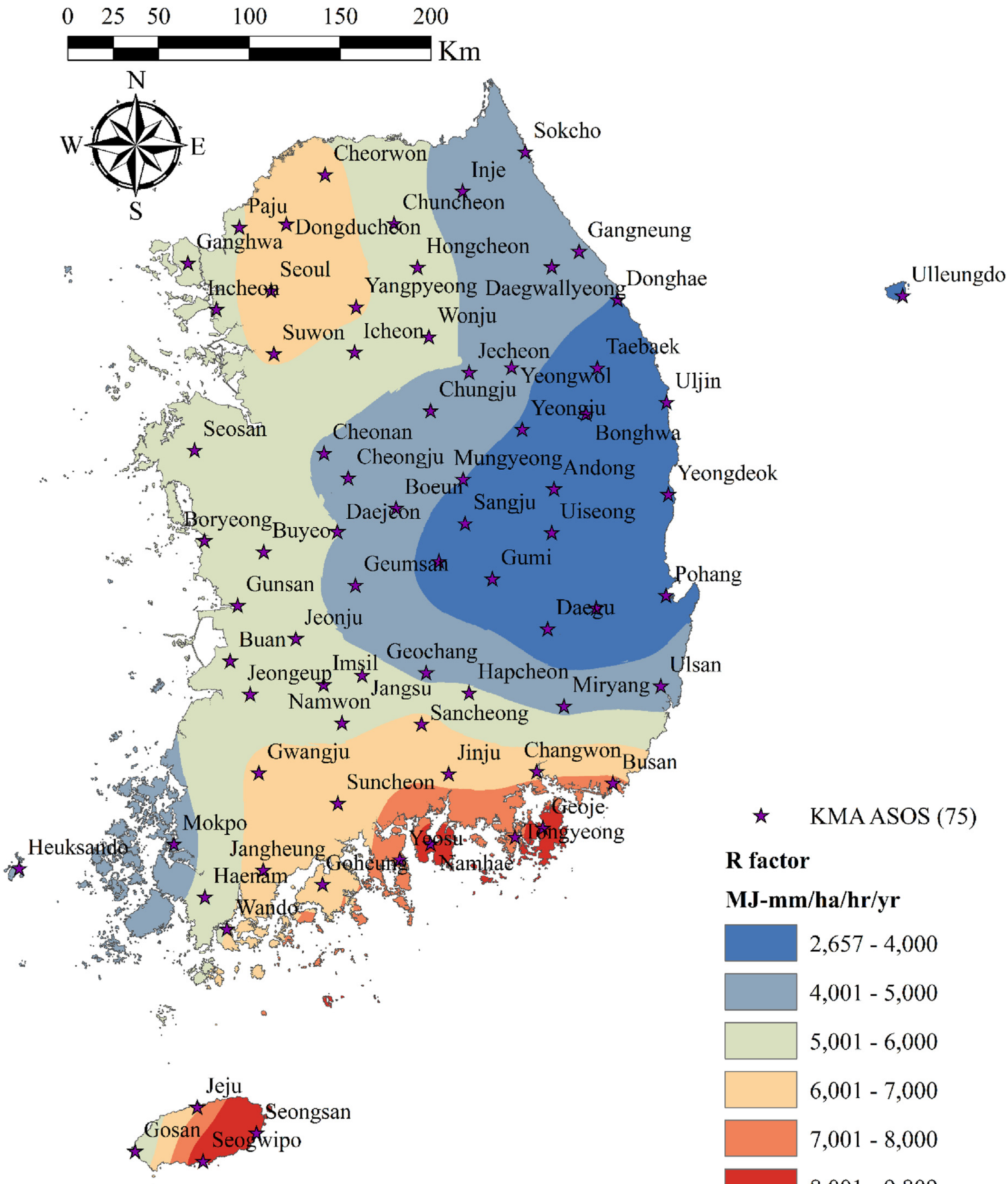

Fig. 4. Iso-erodent map of South Korea showing areas of equal rainfall erosivity based on kriging interpolation of average annual erosivity (2000–2023).

variation in rainfall erosivity values as influenced by changes in P_{md} and P_{10}. The actual monthly rainfall erosivity values, calculated based on the USLE, were overlaid as points on the contour plot. The results indicate that the contour lines closely align with the actual trends in rainfall erosivity, demonstrating that the R9 regression model effectively captures the relationship between rainfall erosivity and its primary influencing variables. However, some points were observed to be located at lower values on the contour plot compared to the actual erosivity values. This suggests that additional complex factors—such as the temporal distribution of rainfall, which was not fully accounted for by P_{md} or P_{10} alone—might also influenced rainfall erosivity. Furthermore, except in cases where P_{md} is below 10 mm, P_{10} was consistently greater than P_{md}, resulting in the actual distribution of erosivity values being positioned below the $y = x$ line. Additionally, P_{10} generally did not exceed ten times the value of P_{md}, which indicated that the actual rainfall erosivity values were distributed within an approximate range of 6°–45° on the contour plot.

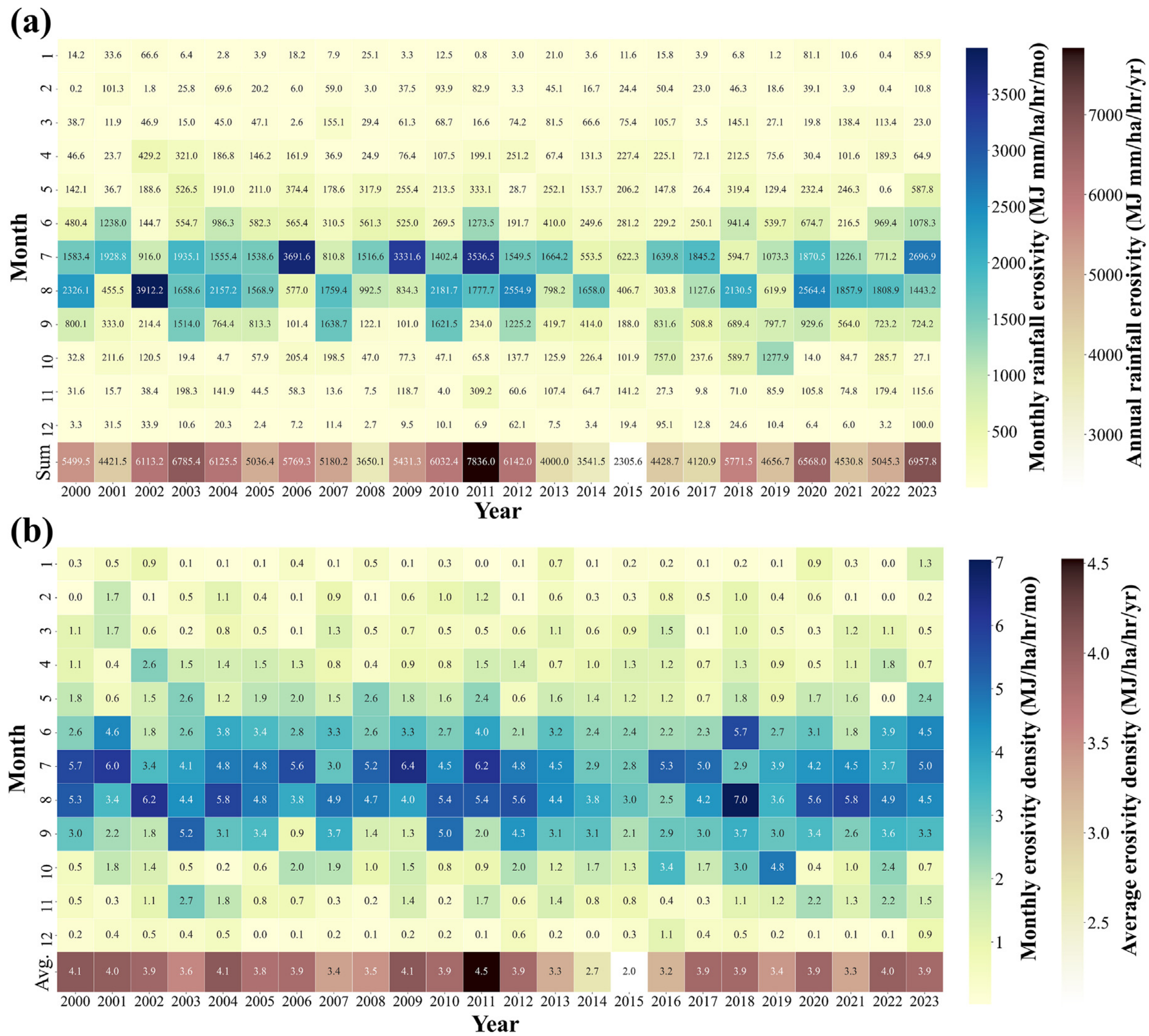

Fig. 5. Heatmap of (a) average monthly rainfall erosivity and (b) monthly erosivity density of 75 ASOS stations for the years 2000–2023.

Table 3
Best-fit regression models and calibrated regression equations for monthly rainfall erosivity factor prediction by parameter combination.

Type	Parameter	References	Model	Regression equation
R1	P_m	Huang et al. (1992); Wu (1994)	Power	$R = 0.158 \cdot P_m^{1.596}$
R2	MFI_m	Arnoldus (1977); Pontes et al. (2017)	Exponential plus linear	$R = 1083.096 - 1094.503 \cdot 0.987^{MFI_m} + 14.033 \cdot MFI_m$
R3	P_{md}	Diodato (2006)	Hoerl curve	$R = 0.120 \cdot 0.997^{P_{md}} \cdot P_{md}^{2.091}$
R4	P_{10}	De Santos Loureiro and de Azevedo Coutinho (2001); Capolongo et al. (2008)	Hoerl curve	$R = 0.130 \cdot 0.999^{P_{10}} \cdot P_{10}^{1.674}$
R5	P_m, P_{md}	This study	Power (E)	$R = 0.188 \cdot P_m^{1.111} \cdot P_{md}^{0.571}$
R6	P_m, M	Diodato and Bellocchi (2007); Risal et al. (2016)	Power (E)	$R = 0.065 \cdot P_m^{1.579} \cdot M^{0.498}$
R7	MFI_m, P_{md}	This study	Power (E)	$R = 6.662 \cdot MFI_m^{0.547} \cdot P_{md}^{0.682}$
R8	P_{10}, P_{10d}	Diodato (2004); De Santos Loureiro and de Azevedo Coutinho (2001); Capolongo et al. (2008)	Simplified quadratic	$R = -28.520 + 6.855 \cdot P_{10} - 81.156 \cdot P_{10d} + 0.006 \cdot P_{10}^2 - 7.963 \cdot P_{10d}^2$
R9	P_{10}, P_{md}	This study	Power (E)	$R = 0.280 \cdot P_{10}^{1.075} \cdot P_{md}^{0.545}$
R10	P_m, P_{10d}	This study	Simplified quadratic	$R = -109.928 + 6.047 \cdot P_m - 69.424 \cdot P_{10d} + 0.007 \cdot P_m^2 - 8.402 \cdot P_{10d}^2$

*Note: P_m, monthly rainfall amount; MFI_m, modified Fournier index of month; P_{md}, monthly maximum daily rainfall amount; P_{10}, monthly rainfall amount for days rainfall $\geq$10 mm; P_{10d}, monthly number of days with rainfall $\geq$10 mm; M, order of month.

3.3. WREC tool

The main page layout and key components of the WREC tool are shown in Fig. 8. At the top of the main page, a banner displays the main visual image of the WREC tool, along with buttons (1, 2) that link to subpages containing explanations of the USLE method and

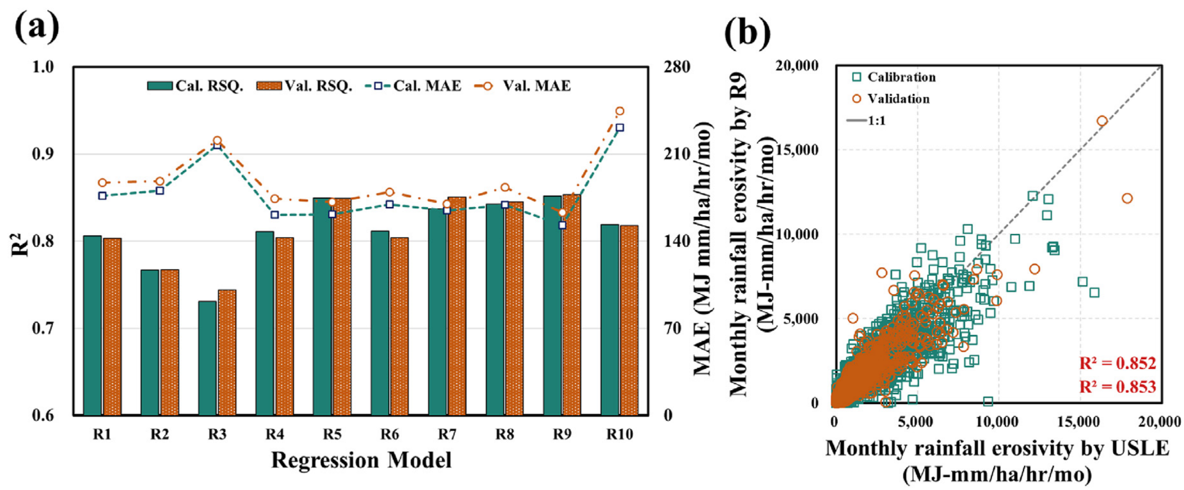

Fig. 6. (a) Comparison of prediction accuracy by regression models and (b) scatter diagrams of actual (USLE-based) and estimated (R9 model-based) monthly rainfall erosivity.

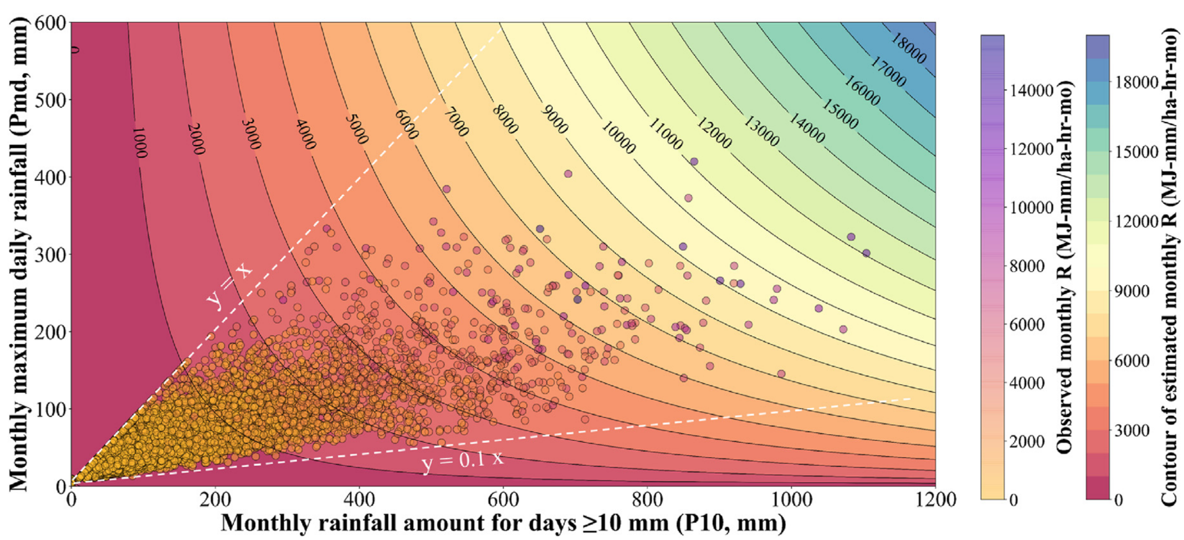

Fig. 7. Contour plot of estimated monthly rainfall erosivity based on the regression model and actual monthly rainfall erosivity values. Regression model R9: contour surface, actual monthly rainfall erosivity: points.

R-factor. Users can interact with box (3) in the central area of the page to select different modules for rainfall erosivity estimation. In section (4), a brief description of the analysis procedures for each module is provided along with links to download the sample input files. Footer section (5) includes important links related to the R-factor.

Users can select an appropriate analysis module based on the available rainfall data. The results, including the monthly and annual rainfall erosivity estimates, are displayed in the tables and graphs of the respective subpages. Additionally, users can download these results as text files (*. txt).

3.3.1. DB-based module

The DB-based module (Fig. 9(a)) provides users with monthly rainfall erosivity for selected weather stations and analysis periods by utilizing the rainfall erosivity database from the 75 weather stations across South Korea. Users can select an administrative region, which will display weather stations within that area. By entering the desired start and end years for the analysis, users can view the rainfall erosivity values for a specified period. This module does not require any additional input data. It presents the annual rainfall and rainfall erosivity values in a table, and the monthly rainfall and erosivity values in a graph.

3.3.2. USLE-based module

The USLE-based module (Fig. 9(b)) processes 1-min interval rainfall data accumulated over one day and provides rainfall erosivity estimates based on the USLE method. Users must upload the input file in a text format (*. txt), with arbitrary file names. The file should contain datetime entries in the “YYYY-MM-DD H:mm” format and corresponding 1-min interval rainfall amounts, with columns separated by tabs. The module handles missing data without additional preprocessing. After uploading the file and clicking the "Calculate R Factor" button, users can view monthly R-values in both table and graph formats. In addition, users can download a result file that includes the $I_{30\ max}$ and EI_{30} values for each rainfall event.

3.3.3. Regression-based module

The regression-based module (Fig. 9(c)) estimates the rainfall

Fig. 8. Main page layout and key components of the WREC tool web interface.

erosivity using a regression model based on daily rainfall data. The input file should include the year (YYYY), month (MM), day (DD), and daily rainfall amount in sequence, with each column separated by a tab. This module allows users to apply not only the monthly rainfall erosivity estimation model derived in this study ($R = 0.280 \cdot P_{10}{}^{1.075} \cdot P_{md}{}^{0.545}$) but also other monthly rainfall erosivity estimation models proposed in studies from various countries (Jung et al., 1983; Plangoen and Babel, 2014; Lee & Lin, 2015; Pontes et al., 2017). After uploading the input file and selecting the desired regression model, users can click the "Calculate R Factor" button to view the monthly rainfall and erosivity value in both table and graph formats.

4. Discussions

Table 4 presents the estimated average R-factors for 2000–2023 at 75 meteorological stations using both the newly proposed regression equation (M1) from this study and existing regression equations (M2–M6) from previous domestic studies. The simulation performances of each regression equation were also compared.

Although some previous studies have proposed regression equations for specific meteorological stations, this study only considered equations applicable across all stations in South Korea to evaluate the suitability of large-scale estimations of rainfall erosivity. M1–M3 estimated the monthly rainfall erosivity based on daily and monthly rainfall data, and M4–M6 estimated the annual rainfall erosivity based on annual rainfall data.

The predictive performance of each regression equation was assessed by comparing the estimated average annual rainfall erosivity with those calculated using the USLE method based on 1-min rainfall data. The results indicated that the predictive performances were ranked as follows: M1 > M3 > M6 > M4 > M5 > M2. M1, M3, and M6 are regression equations derived using rainfall erosivity values calculated from high-resolution rainfall data at 1- and 10-min intervals. These equations exhibited relatively small absolute errors compared with the rainfall erosivity values calculated using the USLE method, as shown by the spatial distribution of rainfall erosivity in Fig. 10. In contrast, M2, M4, and M5 significantly underestimated the rainfall erosivity values.

The underestimation by M2 and M4 was likely because these equations, proposed by Jung et al. (1983), were derived from rainfall data collected before 1980 and therefore do not reflect recent rainfall characteristics. Additionally, the average data period used in this study was only approximately 10 years, which is not sufficiently long (≥20 years) to ensure the accuracy of the developed

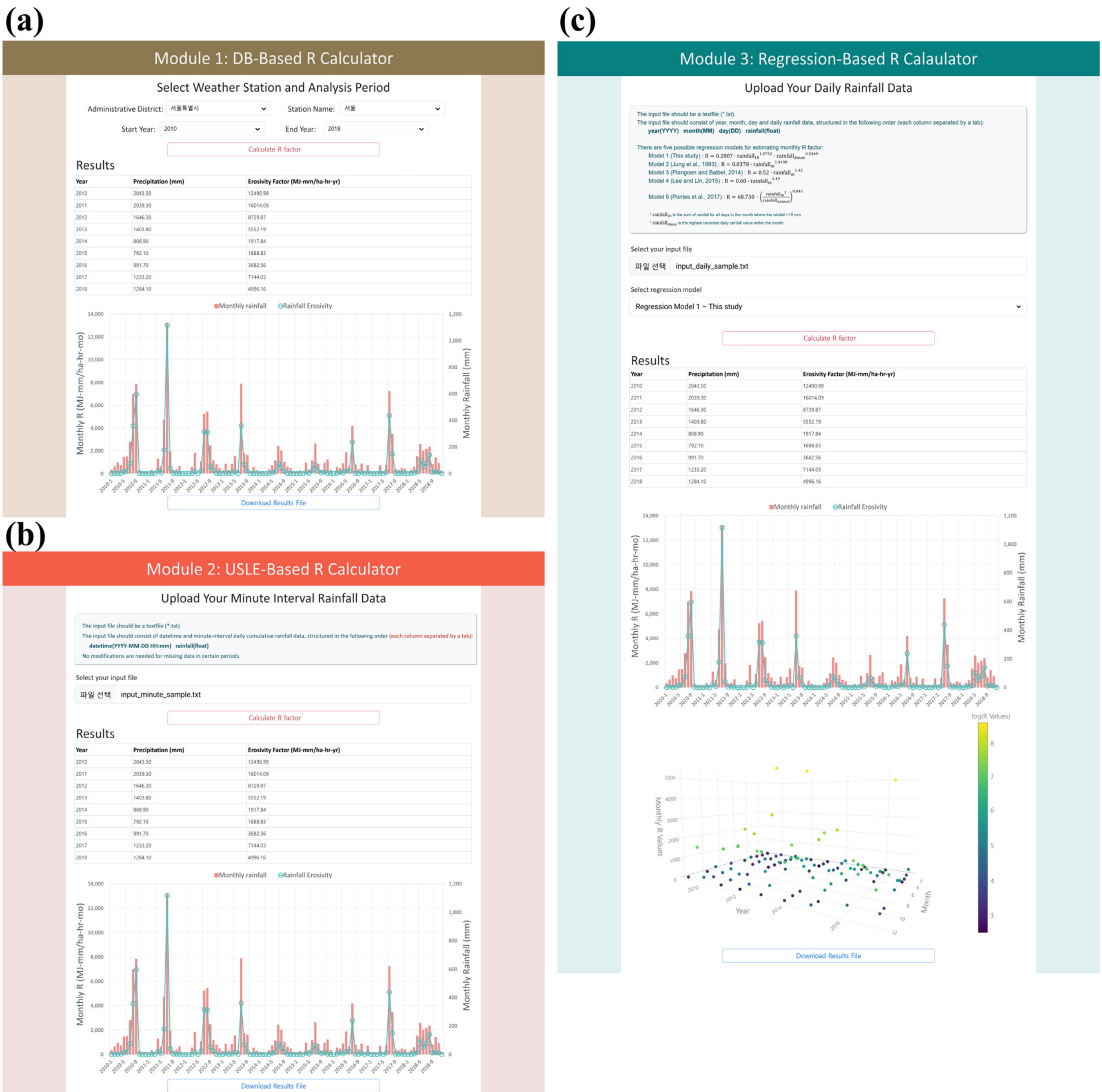

Fig. 9. Results interface for each module in WREC: (a) Module 1, (b) Module 2, and (c) Module 3.

Table 4
Previous and newly-developed regression equations for estimating rainfall erosivity in South Korea and the prediction accuracy of each equation.

Type	References	Data (period)	Regression equation	R factor (MJ-mm/ha/hr/yr)	R^2	MAE (MJ-mm/ha/hr/yr)
M1	This study	1 min (2000–2023)	$R_m = 0.280 \cdot P_{10}{}^{1.075} \cdot P_{md}{}^{0.545}$	5,530	0.866	486.9
M2	Jung et al. (1983)	Pluviograph (1961–1980)	$R_m = 0.0378 \cdot P_m{}^{1.419}$	488	0.826	4,764.0
M3	Risal et al. (2016)	10 min (1997–2014)	$R_m = 0.062 \cdot P_m{}^{1.593} \cdot M^{0.479}$	5,411	0.791	624.5
M4	Jung et al. (1983)	Pluviograph (1961–1980)	$R_a = 0.0115 \cdot P_a{}^{1.4947}$	568	0.793	4,684.4
M5	Lee et al. (2008)	R factors from previous studies (1960–2001)	$R_a = 0.535 \cdot P_a - 230.325$	494	0.787	4,757.9
M6	Lee and Heo (2011)	10 min (1971–1999)	$R_a = 5.86 \cdot P_a - 1715$	6,224	0.787	1,004.3

*Note: R_m, monthly rainfall erosivity factor; P_{10}, monthly rainfall amount for days rainfall ≥10 mm; P_{md}, monthly maximum daily rainfall amount; M, order of month; R_a, annual rainfall erosivity factor; P_a, annual rainfall amount.

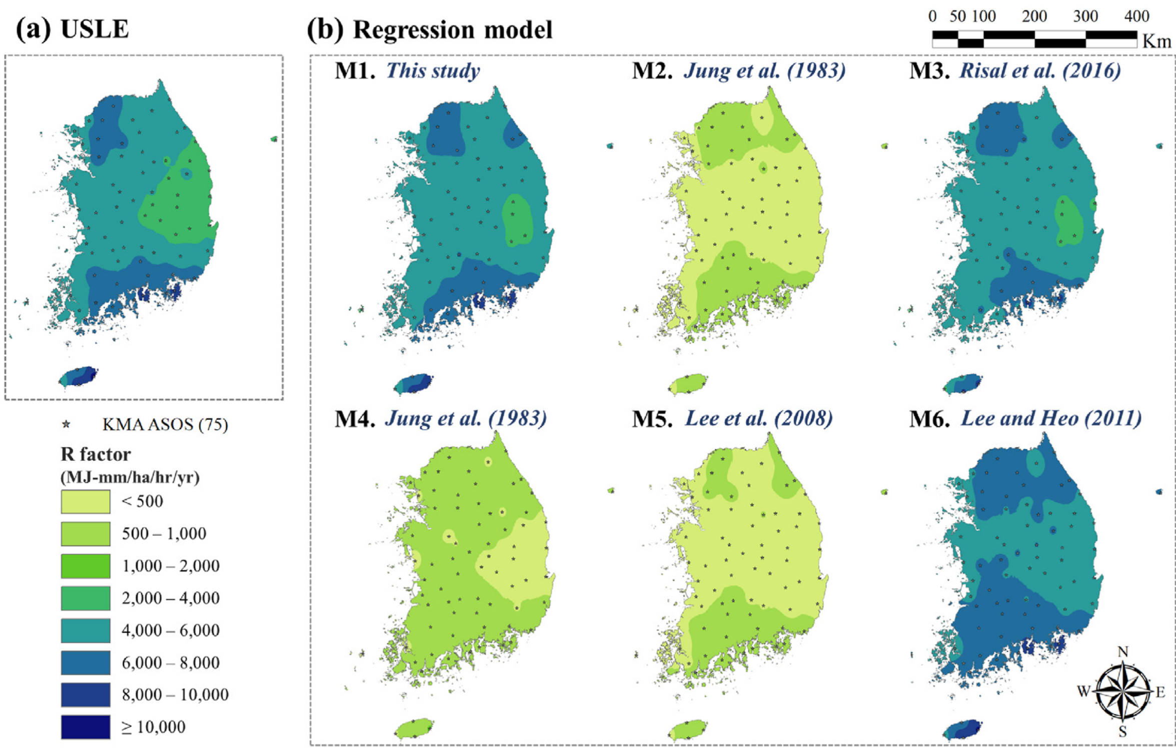

Fig. 10. Maps of average annual rainfall erosivity based on (a) USLE (1-min rainfall data) and (b) previous and newly developed regression equations.

regression equations. Furthermore, M5, proposed by Lee et al. (2008), was developed by synthesizing rainfall erosivity values from previous studies to establish a new relationship between the annual precipitation and rainfall erosivity. However, the 122 rainfall erosivity values used to develop this regression equation originated from various sources with inconsistent temporal resolutions and data periods, which likely contributes to its lower accuracy.

Among M1, M3, and M6, the newly proposed M1 showed the highest performance in both the R^2 and MAE metrics (R^2: 0.866, MAE: 486.9 MJ-mm/ha/hr/yr). Additionally, the spatial distribution of rainfall erosivity estimated by M1 closely matched that estimated by the USLE. This superior performance of M1 is likely due to its incorporation of variables accounting for extreme rainfall events, such as P_{10} (monthly rainfall amount for days with rainfall $\geq$10 mm) and P_{md} (monthly maximum daily precipitation). By incorporating these extreme rainfall variables, M1 more accurately reflected seasonal variability in rainfall erosivity in South Korea, where storms and heavy showers are concentrated in the summer. Consequently, M1 reduced the error in estimating the annual rainfall erosivity by accurately reflecting monthly variations.

Fig. 10 illustrates that despite both M1 and M3 exhibiting similar spatial distributions of rainfall erosivity, M1 more accurately simulated the spatial distribution in regions with high erosivity values (>6,000 MJ-mm/ha/hr/yr) located in the northwestern and southeastern areas. Conversely, M6, which estimated rainfall erosivity using linear regression with annual precipitation, tended to overestimate rainfall erosivity and exhibited a spatial distribution significantly different from that obtained using the USLE method. This suggests that a linear regression model with limited parameters may be insufficient to fully capture the relationship between rainfall and rainfall erosivity.

Furthermore, M1-M3, which estimate monthly rainfall erosivity values and aggregated them to derive annual values, better captured the spatial variability of rainfall erosivity compared with M4–M6, which rely solely on annual precipitation data. This demonstrates that monthly rainfall models offer superior performance in simulating the spatial distribution of rainfall erosivity compared with annual models.

5. Conclusions

Soil erosion remains a significant global challenge with substantial impacts on agricultural productivity, environmental health, and ecosystem stability. The R-factor plays a crucial role in predicting soil erosion by accounting for kinetic energy and rainfall intensity. An accurate estimation of this factor is essential for effective soil erosion management and environmental conservation. Given the complexity and labor-intensive nature of calculating rainfall erosivity using the USLE method, as well as the low accuracy of simplified estimation methods, our objectives were centered on enhancing both the accuracy and accessibility of rainfall erosivity estimations, regardless of the temporal resolution of the rainfall data.

This study aims to address the challenges associated with calculating rainfall erosivity by developing a comprehensive algorithm for processing minute-interval rainfall data, creating a database of erosivity values in South Korea, and deriving an optimal regression model for estimating monthly rainfall erosivity from daily rainfall data. The result is the WREC tool, which enables the accurate determination of rainfall erosivity using minute-interval data based on the USLE method and provides estimations of rainfall erosivity from daily data through simplified estimation methods.

The WREC tool comprises several modules designed to meet diverse needs. The DB-based module provides rapid access to historical rainfall erosivity values, facilitating regional assessments and historical comparisons without additional data inputs. The USLE-based module utilizes high-resolution rainfall data to determine rainfall erosivity for individual rainfall events using the USLE method, and it handles complex calculations swiftly and accurately. The regression-based module caters to users who use daily rainfall data, and offer a variety of regression models, including the newly proposed model and other established models. This module is particularly useful for regions without high-resolution data and for assessing future changes due to climate change.

The WREC tool presents several advantages: it is versatile, supports both detailed calculations from high-resolution data and simplified estimations from daily data, and adapts to different data availability scenarios. As a web-based platform, it ensures accessibility to any location, making it convenient for researchers, policymakers, and land managers to perform rainfall erosivity calculations. Additionally, its user-friendly interface allows straightforward data input, module selection, and result presentation in both tabular and graphical formats, thereby enhancing its usability and efficiency.

The updated R-factors, derived from over 20 years of high-resolution (1-min interval) rainfall data, could be adopted as a new standard for estimating soil erosion in South Korea. Additionally, the monthly rainfall erosivity estimation regression model developed in this study is valuable for predicting future soil erosion under climate change scenarios. This will aid policymakers in anticipating environmental changes and formulating effective soil erosion prevention strategies.

The expansion of the WREC tool to include a broader range of climatic and topographic conditions on a global scale, as well as the regionalization of the regression models to reflect specific local factors, will significantly enhance our ability to address the global challenges of soil erosion. Furthermore, integrating climate change scenarios (e.g., Shared Socioeconomic Pathways scenarios) into the WREC tool would allow for the projection of future rainfall erosivity and soil erosion risks. This would enable policymakers to assess long-term erosion trends and develop adaptive land management strategies. Additionally, incorporating extreme rainfall event analysis and GIS-based spatial visualization could enhance the tool's capability in identifying vulnerable regions under changing climate conditions. Expanding the model to include soil properties and topographic influences would further improve its applicability for comprehensive soil erosion assessments in the context of climate change.

In addition, future research should focus on refining the R-factor estimation by incorporating the latest rainfall energy equations from RUSLE2, which build upon the conventional USLE methodology. Additionally, beyond simply processing rainfall data at minute or daily intervals, it would be valuable to explore the applicability of rainfall time-scale disaggregation models (e.g., Multiplicative Random Cascade, Neyman-Scott Rectangular Pulse Model). This would enable users to estimate rainfall erosivity values across various time resolutions, such as converting hourly data into minute intervals or monthly data into daily intervals, thus enhancing the accuracy of the rainfall erosivity estimation under different temporal scales.

To further improve the WREC tool's functionality, future studies should focus on expanding the spatial applicability and improving the interfaces and functionalities of each module to provide more effective decision support. Additionally, it is necessary to develop a soil erosion prediction module that incorporates other USLE factors, such as soil and topography. Such advancements are expected to promote sustainable land and water management practices, thereby improving the capacity to effectively respond to environmental changes.

CRediT authorship contribution statement

Sinae Kim: Writing – original draft, Software, Methodology, Investigation, Conceptualization. **Seung-Oh Hur:** Conceptualization, Methodology. **Jihye Kwak:** Resources, Methodology, Data curation. **Jihye Kim:** Methodology, Investigation. **Moon-Seong Kang:** Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by a Rural Development Administration (RDA) grant funded by the South Korean government, Ministry of Agriculture, Food, and Rural Affairs (MAFRA) (Grant No: RS-2024-00400192).

References

Alewell, C., Borrelli, P., Meusburger, K., & Panagos, P. (2019). Using the USLE: Chances, challenges and limitations of soil erosion modelling. *Int. Soil Water Conserv. Res., 7*(3), 203–225. https://doi.org/10.1016/j.iswcr.2019.05.004

Araujo, T. F., Sone, J. S., Sobrinho, T. A., Tarso, P., & Oliveira, S. T. S. de (2018). Assessing rainfall erosivity in a tropical region using free software, in: XIII Brazilian meeting of sediment engineering. I particles in the americas. *Hydro sedimentology in the nexus context for a sustainable society*.

Arnoldus, H. M. J. (1977). Methodology used to determine the maximum potential average annual soil loss due to sheet and rill erosion in Morocco. *FAO Soils Bulletin, 34*, 39–51.

Bonilla, C. A., & Vidal, K. L. (2011). Rainfall erosivity in central Chile. *Journal of Hydrology (Amsterdam), 410*(1–2), 126–133. https://doi.org/10.1016/j.jhydrol.2011.09.022

Borrelli, P., Alewell, C., Alvarez, P., Anache, J. A. A., Baartman, J., Ballabio, C., & Panagos, P. (2021). Soil erosion modelling: A global review and statistical analysis. *Science of the Total Environment, 780*, Article 146494. https://doi.org/10.1016/j.scitotenv.2021.146494

Borrelli, P., Robinson, D. A., Panagos, P., Lugato, E., Yang, J. E., Alewell, C., Wuepper, D., Montanarella, L., & Ballabio, C. (2020). Land use and climate change impacts on global soil erosion by water (2015–2070). *Proceedings of the National Academy of Sciences, 117*(36), 21994–22001. https://doi.org/10.1073/pnas.2001403117

Brown, L. C., & Foster, G. R. (1987). Storm erosivity using idealized intensity distributions. *Transactions of the American Society of Agricultural Engineers, 30*(2), 379–386.

Cao, Y., Hua, L., Tang, Q., Liu, L., & Cai, C. (2023). Evaluation of monthly-scale soil erosion spatio-temporal dynamics and identification of their driving factors in Northeast China. *Ecological Indicators, 150*, Article 110187. https://doi.org/10.1016/j.ecolind.2023.110187

Capolongo, D., Diodato, N., Mannaerts, C., Piccarreta, M., & Strobl, R. O. (2008). Analyzing temporal changes in climate erosivity using a simplified rainfall erosivity model in Basilicata (southern Italy). *Journal of Hydrology, 356*(1–2), 119–130. https://doi.org/10.1016/j.jhydrol.2008.04.002

Cardoso, D. P., Silva, E. M., Avanzi, J. C., Muniz, J. A., Ferreira, D. F., Silva, M. L. N., & Curi, N. (2020). RainfallErosivityFactor: An R package for rainfall erosivity (R-factor) determination. *Catena, 189*, Article 104509. https://doi.org/10.1016/j.catena.2020.104509

Chen, Y., Wei, T., Li, J., Xin, Y., & Ding, M. (2024). Future changes in global rainfall erosivity: Insights from the precipitation changes. *Journal of Hydrology*. https://doi.org/10.1016/j.jhydrol.2024.131435, 131435.

De Santos Loureiro, N., & de Azevedo Coutinho, M. (2001). A new procedure to estimate the RUSLE EI30 index, based on monthly rainfall data and applied to the Algarve region, Portugal. *Journal of Hydrology, 250*(1–4), 12–18. https://doi.org/10.1016/S0022-1694(01)00387-0

De Vente, J., & Poesen, J. (2005). Predicting soil erosion and sediment yield at the basin scale: Scale issues and semi-quantitative models. *Earth-Science Reviews, 71*, 95–125. https://doi.org/10.1016/j.earscirev.2005.02.002

Diodato, N. (2004). Estimating RUSLE's rainfall factor in the part of Italy with a Mediterranean rainfall regime. *Hydrology and Earth System Sciences, 8*(1), 103–107. https://doi.org/10.5194/hess-8-103-2004

Diodato, N. (2006). Predicting RUSLE (Revised Universal Soil Loss Equation) monthly erosivity index from readily available rainfall data in Mediterranean area. *The Environment Times, 26*(1), 63–70.

Diodato, N., & Bellocchi, G. (2007). Estimating monthly (R) USLE climate input in a Mediterranean region using limited data. *Journal of Hydrology, 345*(3–4), 224–236. https://doi.org/10.1016/j.jhydrol.2007.08.008
dos Santos, W. P., Avanzi, J. C., Viola, M. R., Chou, S. C., Acuña-Guzman, S. F., Pontes, L. M., & Curi, N. (2022). Projections of rainfall erosivity in climate change scenarios for the largest watershed within Brazilian territory. *Catena, 213*, Article 106225. https://doi.org/10.1016/j.catena.2022.106225
El-Swaify, S. A., Dangler, E. W., & Shinshiro, L. (1985). *Soil erosion and conversion*. Ankeny, Iowa: Soil Conservation Society of America.
Grauso, S., Diodato, N., & Verrubbi, V. (2010). Calibrating a rainfall erosivity assessment model at regional scale in Mediterranean area. *Environmental Earth Sciences, 60*, 1597–1606. https://doi.org/10.1007/s12665-009-0294-z
Grillakis, M. G., Polykretis, C., & Alexakis, D. D. (2020). Past and projected climate change impacts on rainfall erosivity: Advancing our knowledge for the eastern Mediterranean island of Crete. *Catena, 193*, Article 104625. https://doi.org/10.1016/j.catena.2020.104625
Huang, Y., Lu, C., Zheng, T., Fu, Q., & Xu, J. (1992). Study on rainfall erosivity index in southeastern Fujian. *Journal of Soil and Water Conservation, 4*, 1–5. https://doi.org/10.13870/j.cnki.stbcxb.1992.04.001
Hwang, S., Kim, D. H., Shin, S., & Yoo, S. H. (2016). Spatiotemporal uncertainty of rainfall erosivity factor estimated using different methodologies. *J. Korean Soc. Agric. Engineers, 58*(6), 55–69. https://doi.org/10.5389/KSAE.2016.58.6.055
Intergovernmental Panel on Climate Change (IPCC). (2022). Climate change 2022: Impacts, adaptation, and vulnerability. Contribution of working group II to the sixth assessment report of the intergovernmental Panel on climate change. https://doi.org/10.1017/9781009325844.
Jang, C., Shin, Y., Kum, D., Kim, R., Yang, J. E., Kim, S. C., Hwang, S. I., Lim, K. J., Yoon, J. K., Park, Y. S., & Jung, Y. (2015). Assessment of soil loss in South Korea based on land-cover type. *Stochastic Environmental Research and Risk Assessment, 29*, 2127–2141. https://doi.org/10.1007/s00477-015-1027-3
Johannsen, L. L., Schmaltz, E. M., Mitrovits, O., Klik, A., Smoliner, W., Wang, S., & Strauss, P. (2022). An update of the spatial and temporal variability of rainfall erosivity (R-factor) for the main agricultural production zones of Austria. *Catena, 215*, Article 106305. https://doi.org/10.1016/j.catena.2022.106305
Jung, P., Ko, M., Im, J., Um, K., & Choi, D. (1983). Rainfall erosion factor for estimating soil loss. *Korean J. Soil Sci. Fert., 16*(2), 112–118.
Kang, M. S., Cho, J. P., Chun, J. A., & Park, S. W. (2009). Assessment of cell based pollutant loadings in an intensive agricultural watershed. *J. Korean Soc. Agric. Engineers, 51*(5), 87–94. https://doi.org/10.5389/KSAE.2009.51.5.087
Kang, M. W., Kim, D. J., Lim, K., & Lee, S. S. (2021). Rainfall erosivity factor of Korean soils estimated by using USLE under climate change. *Korean J. Soil Sci. Fert., 54*(3), 265–275. https://doi.org/10.7745/KJSSF.2021.54.3.265
Kang, M. S., Park, S. W., Im, S. J., & Kim, H. K. (2003). Computing the half-month rainfall–runoff erosivity factor for RUSLE. *J. Korean Soc. Agric. Engineers, 45*(3), 29–40.
Lee, J. H., & Heo, J. H. (2011). Evaluation of estimation methods for rainfall erosivity based on annual precipitation in Korea. *Journal of Hydrology, 409*(1–2), 30–48. https://doi.org/10.1016/j.jhydrol.2011.07.031
Lee, J. H., Jung, Y. H., & Heo, J. H. (2008). A study on estimation of rainfall erosivity in RUSLE. In *Proceedings of the Korea water Resources association conference* (pp. 1324–1328). Korea Water Resources Association.
Lee, M. H., & Lin, H. H. (2015). Evaluation of annual rainfall erosivity index based on daily, monthly, and annual precipitation data of rainfall station network in Southern Taiwan. *Int. J. Distrib. Sens. Netw. Int. J. Distributed Sensor Networks, 11*(6), Article 214708. https://doi.org/10.1155/2015/214708
Lee, J. H., Shin, J. Y., & Heo, J. H. (2011). Evaluation of rainfall erosivity in Korea using different kinetic energy equations. *Korean J. Soil Sci. Fert., 44*(3), 337–343. https://doi.org/10.7745/KJSSF.2011.44.3.337
Lin, C. Y., Lin, W. T., & Chou, W. C. (2002). Soil erosion prediction and sediment yield estimation: The Taiwan experience. *Soil Till. Res., 68*, 143–152. https://doi.org/10.1016/S0167-1987(02)00114-9
Lo, A., El-Swaify, S. A., Dangler, E. W., & Shinshiro, L. (1985). Effectiveness of EI30 as an erosivity index in Hawaii. In S. A. El-Swaify, W. C. Moldenhauer, & A. Lo (Eds.), *Soil erosion and conservation* (pp. 384–392). Soil Conservation Society of America, Ankeny.
Marcinkowski, P., & Bagdasaryan, V. (2024). Monthly and annual rainfall erosivity in Poland: An empirical model including winter snowfall effect. *J. Hydrol. Reg. Stud., 56*, Article 102076. https://doi.org/10.1016/j.ejrh.2024.102076
Meena, N. K., Gautam, R., Tiwari, P., & Sharma, P. (2017). Nutrient losses in soil due to erosion. *Journal of Pharmacognosy and Phytochemistry, 6*(6S), 1009–1011.
Ministry of Environment (ME). (2013). *Comprehensive plan for top soil conservation*. Sejong, Korea: Ministry of Environment.
Ministy of the Interior and Safety (MOIS). (2023). *Statistical yearbook of natural disaster 2022*.
Mondal, A., Khare, D., & Kundu, S. (2016). A comparative study of soil erosion modelling by MMF, USLE and RUSLE. *Geocarto International, 33*(1), 89–103. https://doi.org/10.1080/10106049.2016.1232313
Moreira, M. C., Pruski, F. F., Oliveira, T. E. C. D., Pinto, F. D. A. D. C., & Silva, D. D. D. (2008). NetErosividade MG: rainfall erosivity for Minas Gerais State, Brazil-NetErosividade MG: erosividade da chuva em Minas Gerais. *Rev. Bras. Ciênc. Solo, 32*(3), 1349–1353. https://doi.org/10.1590/S0100-06832008000300042
Mullan, D. (2013). Soil erosion under the impacts of future climate change: Assessing the statistical significance of future changes and the potential on-site and off-site problems. *Catena, 109*, 234–246. https://doi.org/10.1016/j.catena.2013.03.007
OCED. (2008). *Environmental performance of agriculture in OECD countries since 1990*. OECD.
Panagos, P., Ballabio, C., Borrelli, P., Meusburger, K., Klik, A., Rousseva, S., Tadić, M. P., Michaelides, S., Hrabalíková, M., & Olsen, P. (2015). Rainfall erosivity in europe. *Science of the Total Environment, 511*, 801–814. https://doi.org/10.1016/j.scitotenv.2015.01.008
Panagos, P., Borrelli, P., Matthews, F., Liakos, L., Bezak, N., Diodato, N., & Ballabio, C. (2022). Global rainfall erosivity projections for 2050 and 2070. *Journal of Hydrology, 610*, Article 127865. https://doi.org/10.1016/j.jhydrol.2022.127865
Pandey, S., Kumar, P., Zlatic, M., Nautiyal, R., & Panwar, V. P. (2021). Recent advances in assessment of soil erosion vulnerability in a watershed. *Int. Soil Water Conserv. Res., 9*(3), 305–318. https://doi.org/10.1016/j.iswcr.2021.03.001
Park, J. H., Pyeon, J. G., & Kim, G. I. (2000). A study of distribution of rainfall erosivity in USLE/RUSLE for estimation of soil loss. *Journal of Korea Water Resources Association, 33*(5), 603–610.
Park, C. W., Sonn, Y. K., Hyun, B. K., Song, K. C., Chun, H. C., Moon, Y. H., & Yun, S. G. (2011). The redetermination of USLE rainfall erosion factor for estimation of soil loss at Korea. *Korean J. Soil Sci. Fert., 44*(6), 977–982. https://doi.org/10.7745/KJSSF.2011.44.6.977
Polykretis, C., Alexakis, D. D., Grillakis, M. G., & Manoudakis, S. (2020). Assessment of intra-annual and inter-annual variabilities of soil erosion in Crete Island (Greece) by incorporating the Dynamic "Nature" of R and C-Factors in RUSLE modeling. *Remote Sensing, 12*(15), 2439. https://doi.org/10.3390/rs12152439
Pontes, L. M., Silva, M. L. N., Bispo, D. F. A., Avalos, F. A. P., Oliveira, M. S. D., & Rocha, H. R. D. (2017). Spatial distribution of annual and monthly rainfall erosivity in the Jaguarí River Basin. *Rev. Bras. Ciênc. Solo, 41*. https://doi.org/10.1590/18069657rbcs20160407
Prasannakumar, V., Shiny, R., Geetha, N., & Vijith, H. (2011). Spatial prediction of soil erosion risk by remote sensing, gis and RUSLE approach: A case study of siruvani river watershed in attapady valley, Kerala, India. *Environmental Earth Sciences, 64*(4), 965–972. https://doi.org/10.1007/s12665-011-0913-3
Renard, K. G., & Freimund, J. R. (1994). Using monthly precipitation data to estimate the R-factor in the revised USLE. *Journal of Hydrology, 157*, 287–306. https://doi.org/10.1016/0022-1694(94)90110-4
Rickson, R. J. (2014). Can control of soil erosion mitigate water pollution by sediments? *Science of the Total Environment, 468–469*, 1187–1197. https://doi.org/10.1016/j.scitotenv.2013.05.057
Risal, A., Bhattarai, R., Kum, D., Park, Y. S., Yang, J. E., & Lim, K. J. (2016). Application of web ERosivity module (WERM) for estimation of annual and monthly R factor in Korea. *Catena, 147*, 225–237. https://doi.org/10.1016/j.catena.2016.07.017
Schmidt, S., Alewell, C., & Meusburger, K. (2019). Monthly RUSLE soil erosion risk of Swiss grasslands. *Journal of Maps, 15*(2), 247–256. https://doi.org/10.1080/17445647.2019.1585980
Sepaskhah, A. R., & Sarkhosh, P. (2005). Estimating storm erosion index in southern region of IR Iran. Iran. *J. Sci. Technol. Trans. B Eng., 29*, 357–363.
Shin, J. Y., Kim, T., Heo, J. H., & Lee, J. H. (2019). Spatial and temporal variations in rainfall erosivity and erosivity density in South Korea. *Catena, 176*, 125–144. https://doi.org/10.1016/j.catena.2019.01.005
Wang, L., Li, Y., Gan, Y., Zhao, L., Qin, W., & Ding, L. (2024). Rainfall erosivity index for monitoring global soil erosion. *Catena, 234*, Article 107593. https://doi.org/10.1016/j.catena.2023.107593
Wang, W., Yin, S., Yu, J., He, Z., & Xie, Y. (2023). Long-term trends of precipitation and erosivity over Northeast China during 1961–2020. *Int. Soil Water Conserv. Res., 11*(4), 743–754. https://doi.org/10.1016/j.iswcr.2023.04.002
Wischmeier, W. H., & Smith, D. D. (1978). *Predicting rainfall erosion losses: A guide to conservation planning*. Department of Agriculture, Science and Education Administration. No. 537.
Wu, S. (1994). Calculation model of the sediment delivery ratio on sloping land in gullied rolling loess region. *Soil Water Conserv. China, 4*, 12–13.
Xu, X., Yun, X., Tang, Q., Cui, H., Wang, J., Zhang, L., & Chen, D. (2023). Projected seasonal changes in future rainfall erosivity over the Lancang-Mekong River basin under the CMIP6 scenarios. *Journal of Hydrology, 620*, Article 129444. https://doi.org/10.1016/j.jhydrol.2023.129444
Yin, S., Nearing, M. A., Borrelli, P., & Xue, X. (2017). Rainfall erosivity: An overview of methodologies and applications. *Vadose Zone Journal, 16*(12), 1–16. https://doi.org/10.2136/vzj2017.06.0131
Yin, S., Xie, Y., Liu, B., & Nearing, M. A. (2015). Rainfall erosivity estimation based on rainfall data collected over a range of temporal resolutions. *Hydrology and Earth System Sciences, 19*, 4113–4126. https://doi.org/10.5194/hess-19-4113-2015
Yu, B., & Rosewell, J. C. (1996). Technical notes: A robust estimator of the R-factor for the universal soil loss equation. *Transactions of the American Society of Agricultural Engineers, 39*(2), 559–561. https://doi.org/10.13031/2013.27535